

Satellite to Building Level Mapping for Urban Heat and Energy Assessment

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Abstract—India’s building sector consumes roughly one-third of national electricity, yet building-level energy-use data remain largely unavailable for the cities most relevant to policy. This paper presents a scalable, physics-informed GeoAI pipeline that maps annual energy-use intensity (EUI) from open satellite and building-footprint data across four climatically distinct Indian cities—Delhi NCR (Composite zone), Ahmedabad (Hot & Dry), Kolkata (Warm & Humid), and Bengaluru (Temperate)—covering all active Energy Conservation Building Code (ECBC) zones relevant to northern and peninsular India. For each city a two-stage workflow is executed: a simplified 1R-1C resistance-capacitance (RC) thermal simulator generates physics-consistent annual energy targets for 5,000 buildings drawn from the Google Open Buildings dataset, and a stacking ensemble surrogate (Random Forest, XGBoost, LightGBM, Ridge meta-learner) is subsequently trained on these targets using 21 geospatial features grouped into three interpretable classes—satellite spectral indices (Class 1), building-morphology priors (Class 2), and city-level climate variables (Class 3). A spatial train-test split on a latitudinal threshold prevents spatial autocorrelation leakage. Per-city surrogate accuracy ranges from $R^2 = 0.889$ (Ahmedabad) to $R^2 = 0.969$ (Kolkata), and a pooled cross-city model achieves $R^2 = 0.971$ on 4,000 held-out buildings. A Leave-One-City-Out (LOCO) cross-city generalisation experiment further demonstrates that models trained on three zones transfer effectively to Hot & Dry ($R^2 = 0.900$), Composite ($R^2 = 0.861$), and Warm & Humid ($R^2 = 0.819$) targets, while the Temperate zone exhibits the sharpest degradation ($R^2 = 0.628$), indicating that low-HVAC-demand climates require dedicated calibration data. Feature ablation reveals that satellite spectral indices alone explain negligible variance (pooled $R^2 = 0.002$), whereas adding building-morphology features raises pooled R^2 to 0.858, and incorporating city-level climate variables further improves it to 0.877. At city scale, validated surrogates are applied to approximately 100,000 real buildings per city in under one second, achieving speedups of 2,942–18,193 $\times$ relative to the underlying RC simulator. The framework provides a reproducible, open-data foundation for national-scale EUI mapping to support ECBC compliance monitoring and urban-retrofit planning.

Index Terms—GeoAI, building energy modeling, surrogate model, urban energy assessment, Sentinel-2, spectral indices, ECBC climate zones, Leave-One-City-Out, cross-city generalization, India

I. INTRODUCTION AND BACKGROUND

Buildings consume approximately 40% of global final energy and are responsible for nearly a third of worldwide greenhouse gas emissions [1]. In the Indian context these figures carry particular weight: the country’s urban population is projected to exceed 600 million by 2030, and the building

stock is expected to triple in floor area between 2020 and 2050 [2]. Despite this trajectory, city-level building energy data in India remain largely inaccessible. Distribution companies (DISCOMs) aggregate consumption at the feeder or substation level; individual building utility records are seldom publicly available; and the Energy Conservation Building Code (ECBC), though mandatory for new large commercial buildings since 2007, lacks a systematic enforcement and monitoring mechanism.

Three practical bottlenecks constrain city-scale energy assessment in this context. First, building-level metadata—floor count, construction vintage, envelope properties, occupancy rate—are missing for the majority of the urban stock. Second, computational throughput: detailed simulation tools such as EnergyPlus require upward of thirty seconds per building, rendering full-stock runs over hundreds of thousands of records infeasible in interactive planning workflows. Third, measured calibration data against which any model can be validated are scarce. Together these constraints have restricted most published building energy studies in India to small samples or to stylised archetypes [3].

This paper addresses all three bottlenecks through a physics-informed machine learning surrogate that derives its inputs exclusively from open geospatial sources—the Google Open Buildings (GOB) footprint dataset and a cloud-free Sentinel-2 multispectral composite—and generates physics-consistent annual energy targets through a simplified resistance-capacitance (RC) thermal model driven by city-specific Energy Plus Weather (EPW) files.

The study covers four cities that together span four of the five active ECBC climate zones: Delhi NCR (Composite), Ahmedabad (Hot & Dry), Kolkata (Warm & Humid), and Bengaluru (Temperate). Each city has a distinct energy demand signature: Ahmedabad faces extreme cooling loads owing to its desert-adjacent climate ($CDD \approx 3,200$), Delhi experiences both a significant winter heating demand and a long cooling season ($CDD \approx 3,000$), Kolkata operates under persistent warm-humid conditions with minimal heating ($HDD \approx 20$), and Bengaluru’s mild temperate climate produces the lowest overall mechanical-conditioning demand ($CDD \approx 700$). Modeling this diversity within a single consistent pipeline is the central technical challenge.

II. LITERATURE REVIEW

A. Spectral Indices for Urban Built-Up Extraction

The extraction of urban morphological features from multispectral satellite imagery relies on normalised-difference indices that exploit reflectance contrasts between built surfaces, vegetation, water, and bare soil. The Normalised Difference Vegetation Index (NDVI) [4] and Normalised Difference Built-Up Index (NDBI) [5] are the most widely used pair in the energy-relevant literature, the former serving as a proxy for shading and microclimate cooling, the latter as an indicator of impervious surface density and thermal emissivity. The Modified Normalised Difference Water Index (MNDWI) [8], Urban Index (UI) [9], and Bare Soil Index (BSI) [10] extend this spectral vocabulary to water bodies, urban heat islands, and exposed mineral surfaces respectively.

More recent work has focused on improving the separability between built-up classes and spectrally similar land covers. The Enhanced Normalised Difference Impervious Surfaces Index (ENDISI) [6], calibrated for Indian cities using a Scaled Difference Index (SDI) value of 5.29, has been shown to outperform earlier indices on Sentinel-2 imagery over Mumbai suburban districts [7]. A systematic evaluation of seven spectral indices on Sentinel-2 imagery reported that NDBI, UI, and ENDISI together provided the highest discrimination accuracy for impervious surface extraction when fused by majority vote [11], a finding that directly motivates the building-type classifier used in the present pipeline.

B. Physics-Informed and Surrogate Building Energy Models

Purely data-driven building energy models trained on metered consumption data have demonstrated impressive accuracy at portfolio scale [13], but they cannot be deployed in data-scarce cities because no calibration labels exist. Physics-based tools such as EnergyPlus and TRNSYS offer physical fidelity but become computationally intractable for stocks exceeding a few thousand buildings. The surrogate modelling paradigm—training a fast statistical approximator on physics-generated labels—bridges this gap and has attracted growing interest in the urban energy literature [14].

Within this paradigm the training label quality is determined by the fidelity of the underlying physics engine. RC (resistance-capacitance) lumped-parameter models, sometimes called simplified ISO 13790 models, have been extensively validated for annual energy estimation in residential and commercial archetypes [3]. They are orders of magnitude faster than multi-zone EnergyPlus simulations and are well suited to stochastic batch generation of training labels across diverse building configurations. The present work employs a 1R-1C hourly RC model driven by real EPW weather sequences, producing heating, cooling, lighting, and appliance energy disaggregated at the building level.

C. GeoAI Pipelines for Urban-Scale Energy Assessment

The term GeoAI denotes the intersection of machine learning with geospatially referenced data, encompassing satellite

imagery, footprint databases, and geographic information systems [12]. Applications to building energy have progressed from simple regression on footprint geometry [15] toward end-to-end pipelines that couple feature extraction from very-high-resolution imagery with physics-informed labelling [14]. Building footprint data quality critically affects downstream model performance: studies that leverage large-scale open footprint sources such as Google Open Buildings or Microsoft Building Footprints (MSFT) have demonstrated feasibility at national scale, but significant heterogeneity in footprint completeness and area accuracy across cities requires explicit handling [17].

Cross-city transferability is an emerging concern: most published surrogates are calibrated on a single city, and very few evaluate whether a model trained on one climate generalises to another without re-calibration. The Leave-One-City-Out (LOCO) evaluation framework, drawn from cross-lingual transfer learning [20], offers a principled test of such transferability and is adopted here for the first time in the Indian building energy context.

III. RESEARCH OBJECTIVES AND CONTRIBUTIONS

A. Objectives

This work pursues four concrete objectives. First, to construct a reproducible, open-data pipeline that extracts 21 geospatial features per building from publicly available satellite imagery and footprint databases for any Indian city for which an ECBC EPW file is available. Second, to generate physics-consistent annual energy labels through a calibrated RC thermal model, removing the dependency on measured utility data. Third, to train and evaluate surrogate models both within each city and across cities, characterising how well the surrogate generalises to unseen climate zones. Fourth, to demonstrate city-scale deployment at approximately 100,000 buildings per city, measuring computational speedup relative to direct RC simulation.

B. Contributions

The principal contributions of this paper are as follows.

- 1) **Multi-city, multi-zone pipeline.** The first open, reproducible GeoAI-surrogate pipeline covering four Indian ECBC climate zones within a single consistent methodology, using Sentinel-2 composites and Google Open Buildings data.
- 2) **Three-class feature taxonomy.** A structured decomposition of 21 input features into satellite spectral indices (Class 1, nine features), building-morphology priors (Class 2, nine features), and city-level climate variables (Class 3, three features), with an ablation study that isolates the contribution of each class.
- 3) **Empirical ablation finding.** Quantitative demonstration that satellite spectral indices alone explain weak or negative variance ($R^2 = -0.549$ to 0.223 per city; pooled $R^2 = 0.002$), while adding building-morphology features raises performance to 0.795 – 0.962 , providing a

definitive empirical basis for designing remote-sensing-based energy pipelines.

- 4) **LOCO cross-city generalisation.** A formal Leave-One-City-Out experiment showing that cross-zone generalisation is high for Hot & Dry ($R^2 = 0.900$), Composite ($R^2 = 0.861$), and Warm & Humid ($R^2 = 0.819$) zones, but significantly degraded for the Temperate zone ($R^2 = 0.628$) where mechanical conditioning demand is low and variable.
- 5) **Urban-scale inference at 100k buildings.** Deployment of validated surrogates to $\approx 100,000$ real GOB buildings per city with prediction intervals, achieving $2,942\text{--}18,193\times$ computational speedup relative to the RC simulator.
- 6) **Carbon and EUI benchmarking.** City-level comparison of predicted EUI against Bureau of Energy Efficiency (BEE) Energy Performance Index (EPI) benchmarks, and grid-emission-factor-weighted CO_2 estimates for each city.

IV. STUDY AREA AND DATA SOURCES

A. City Selection and ECBC Zone Coverage

India’s ECBC defines five climate zones for building code compliance based on Cooling Degree Days (CDD) and Heating Degree Days (HDD) relative to base 18°C . The four cities selected cover zones 1 through 5 as follows: Ahmedabad (Zone 1, Hot & Dry, $\text{CDD} \approx 3,200$), Delhi NCR (Zone 2, Composite, $\text{CDD} \approx 3,000$, $\text{HDD} \approx 180$), Kolkata (Zone 4, Warm & Humid, $\text{CDD} \approx 3,500$, $\text{HDD} \approx 20$), and Bengaluru (Zone 5, Temperate, $\text{CDD} \approx 700$). Zone 3 (Warm & Humid coastal) was not included in this iteration owing to the unavailability of a consistent GOB footprint extract for the selected coastal cities at the time of data collection; this gap is identified as a direction for future work.

The geographic characteristics and key parameters of each city are summarised in Table I. Grid emission factors are sourced from CEA regional grid data [18] and reflect the carbon intensity of electricity supply, which varies from $0.79 \text{ kgCO}_2/\text{kWh}$ for the Southern and Western grids (Bengaluru, Ahmedabad) to $0.94 \text{ kgCO}_2/\text{kWh}$ for the Eastern grid (Kolkata).

TABLE I: City Configuration: Climate and Grid Parameters

City	Zone	Grid	CDD	HDD	RH%	EF
Delhi NCR	Composite	North	3000	180	55	0.82
Ahmedabad	Hot & Dry	West	3200	100	48	0.79
Kolkata	Warm & Humid	East	3500	20	80	0.94
Bengaluru	Temperate	South	700	100	65	0.79

EF = CO_2 emission factor (kgCO_2/kWh). CDD/HDD at base 18°C .

B. Building Footprint Data

Building footprint records are sourced from the Google Open Buildings (GOB) Points v3 dataset, filtered at a confidence threshold of 0.75 to reduce commission errors. Where GOB records are unavailable for a particular city, the fallback

hierarchy is Microsoft Building Footprints $\rightarrow$ OpenStreetMap Overpass API $\rightarrow$ synthetic debug inventory (disabled for production runs). Footprint area, centroid latitude, and centroid longitude are the three geometric attributes extracted per building. Floor count is not available from GOB and is instead estimated probabilistically as described in Section V-B.

C. Satellite Imagery

Sentinel-2 Level-2A Surface Reflectance imagery (ESA Copernicus) is used for spectral feature extraction. For each city a cloud-free median composite was assembled from scenes acquired during the dry season (October–March) of 2024, restricting cloud fraction to below 20% using the Sentinel-2 Scene Classification Layer. Six spectral bands are used: Blue (B2), Green (B3), Red (B4), NIR (B8), SWIR1 (B11), and SWIR2 (B12). All bands are sampled at 10 m spatial resolution. Building-centroid spectral values are extracted by point sampling (not polygon zonal statistics), consistent with the validated Delhi baseline methodology documented in the pipeline metadata.

D. Weather Data

Annual hourly dry-bulb temperature sequences are parsed from city-specific EPW files—Delhi (28.6°N), Ahmedabad (23.0°N), Kolkata (22.5°N), Bengaluru (12.9°N)—to drive the 8,760-hour RC thermal simulation. The per-city latitude is passed explicitly to the solar gain model so that the hourly zenith-angle computation is climatically consistent for each city.

V. METHODOLOGY

A. Pipeline Overview

The pipeline is structured as four sequential scripts executed per city and one multi-city aggregation step, as illustrated conceptually in Fig. 1.

A fixed random seed (42) and a per-city `numpy` Generator are propagated through all stochastic steps to ensure reproducibility. Intermediate outputs are checkpointed as CSV and Joblib files so that individual pipeline stages can be re-run without repeating upstream computation.

B. Feature Extraction

For each building a 21-dimensional feature vector is assembled from three source classes, as described in Table II.

1) *Class 1: Satellite Spectral Features (9 features):* Six normalised-difference indices are computed directly from Sentinel-2 centroid reflectances. Let $\epsilon = 10^{-6}$ denote a small regularisation constant throughout.

Figures/pipeline_overview.png

Fig. 1: End-to-end pipeline. Script 01 downloads and filters building footprints; Script 02 extracts 21 geospatial features; Script 03 runs the RC physics simulator; Script 04 trains and evaluates surrogate models including ablation and LOCO; Script 05 deploys surrogates to $\approx 100k$ buildings per city.

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}} + \epsilon} \quad (1)$$

$$\text{NDBI} = \frac{\rho_{\text{SWIR1}} - \rho_{\text{NIR}}}{\rho_{\text{SWIR1}} + \rho_{\text{NIR}} + \epsilon} \quad (2)$$

$$\text{ENDISI} = \frac{\rho_{\text{Blue}} - \alpha}{\rho_{\text{Blue}} + \alpha + \epsilon}, \quad \alpha = \frac{\rho_{\text{Green}} + \rho_{\text{NIR}}}{2\rho_{\text{SWIR1}}} \quad (3)$$

$$\text{MNDWI} = \frac{\rho_{\text{Green}} - \rho_{\text{SWIR1}}}{\rho_{\text{Green}} + \rho_{\text{SWIR1}} + \epsilon} \quad (4)$$

$$\text{UI} = \frac{\rho_{\text{SWIR2}} - \rho_{\text{NIR}}}{\rho_{\text{SWIR2}} + \rho_{\text{NIR}} + \epsilon} \quad (5)$$

$$\text{BSI} = \frac{(\rho_{\text{SWIR1}} + \rho_{\text{Red}}) - (\rho_{\text{NIR}} + \rho_{\text{Blue}})}{(\rho_{\text{SWIR1}} + \rho_{\text{Red}}) + (\rho_{\text{NIR}} + \rho_{\text{Blue}}) + \epsilon} \quad (6)$$

A normalised mean visible reflectance (brightness_norm) is derived by min-max scaling of $(\rho_{\text{Blue}} + \rho_{\text{Green}} + \rho_{\text{Red}})/3$ across the city footprint. A binary building_type label (0=residential, 1=commercial) is assigned by majority vote on three binary indicators ($\text{NDBI} > 0.15$, $\text{ENDISI} > -0.75$, $\text{UI} > 0$), with a 10% stochastic flip to simulate mixed-use uncertainty. The occupancy_proxy feature scales the continuous occupancy estimate by 1.5 to serve as an internal gains scalar for the RC model.

2) *Class 2: Building-Morphology Prior Features (9 features)*: Where the GOB record supplies a footprint area in

square metres that exceeds 20 m^2 , this real measurement is used directly. Otherwise, a spectral proxy is substituted:

$$A_{\text{footprint}} = \text{clip}(100 + 500 \text{NDBI}, 30, 2000) [\text{m}^2] \quad (7)$$

Floor count is not available from open satellite sources. We estimate it from a city-calibrated probabilistic prior with an NDBI-correlated boost, following the approach described in [16]:

$$N_{\text{floors}} = \text{clip}(f_{\text{base}} + f_{\text{boost}}, 1, 20) \quad (8)$$

where $f_{\text{base}} \sim \text{Categorical}(\{1, 2, 3, 4\}, \{0.20, 0.50, 0.20, 0.10\})$ and $f_{\text{boost}} \sim \text{Uniform}(0, 2)$ when $\text{NDBI} > 0.2$ or $\text{ENDISI} > 0.1$. Building height is set as $h = 3N_{\text{floors}} \text{ m}$, consistent with IS:875 Part 1 typical storey heights. The remaining five Class 2 features—building_age_years, occupancy_rate, insulation_proxy, maintenance_score, wwr_proxy, and hvac_efficiency—are drawn from city-calibrated Gaussian or correlation-based priors and represent the epistemic uncertainty inherent in any remote-sensing-only assessment. These are labelled as physics-support priors in the paper and are not presented as remote-sensing observations.

3) Class 3: City-Level Climate Features (3 features):

Three scalar climate descriptors are assigned at the city level from the ECBC configuration: hdd_city, cdd_city, and avg_rh_pct. Although these features carry no within-city spatial variation, they encode the thermal regime of each city and are essential for cross-city surrogate deployment.

TABLE II: Feature Taxonomy: 21 Input Variables

Feature	Class	Source	Reference
NDVI	C1-Satellite	Sentinel-2	[4]
NDBI	C1-Satellite	Sentinel-2	[5]
ENDISI	C1-Satellite	Sentinel-2	[6], [7]
MNDWI	C1-Satellite	Sentinel-2	[8]
UI	C1-Satellite	Sentinel-2	[9]
BSI	C1-Satellite	Sentinel-2	[10]
brightness_norm	C1-Satellite	Sentinel-2	—
building_type	C1-Satellite	Spectral vote	[11]
occupancy_proxy	C1-Satellite	Derived	—
footprint_area_m2	C2-Morphology	GOB / NDBI proxy	[17]
n_floors	C2-Morphology	Prior + NDBI boost	[16]
height_m	C2-Morphology	$3 \times n_{\text{floors}}$	—
building_age_years	C2-Morphology	Brightness prior	—
occupancy_rate	C2-Morphology	Gaussian prior	—
insulation_proxy	C2-Morphology	$1 - \text{age}/100$	—
maintenance_score	C2-Morphology	$0.9 \times \text{insulation}$	—
wwr_proxy	C2-Morphology	$\text{NDBI} + \text{ENDISI}$	—
hvac_efficiency	C2-Morphology	Brightness prior	—
hdd_city	C3-Climate	EPW / ECBC	—
cdd_city	C3-Climate	EPW / ECBC	—
avg_rh_pct	C3-Climate	EPW / ECBC	—

C. RC Physics Simulator

A 1R-1C (single-resistance, single-capacitance) lumped thermal model is solved at hourly timesteps for each building over a full 8,760-hour annual cycle. The state variable is the mean radiant temperature T_m , and the energy balance is:

$$C_{eff} \frac{dT_m}{dt} = Q_{gains}(t) - H_{total} (T_m(t) - T_{out}(t)) \quad (9)$$

where H_{total} is the aggregate heat transfer coefficient:

$$H_{total} = U_{wall}A_{wall} + U_{roof}A_{roof} + U_{win}A_{win} + H_{inf} \quad (10)$$

Infiltration is modelled as:

$$H_{inf} = \frac{ACH \cdot V_{zone}}{3600} \rho_a c_{p,a} \quad (11)$$

Solar gains through glazing and absorbed roof irradiance are computed from the Declination–Hour–Angle model at the city latitude ϕ :

$$\cos \theta_z = \sin \phi \sin \delta + \cos \phi \cos \delta \cos \omega \quad (12)$$

$$Q_{solar}(t) = \max(\cos \theta_z, 0) \cdot \frac{G_{ann}}{4.38} [A_{win} \cdot SHGC \cdot \eta_{shade} + A_{roof} \cdot \alpha_{roof} \cdot 0.04] \quad (13)$$

where G_{ann} is the annual global horizontal irradiance for the city (from EPW), $4.38 \text{ kWh m}^{-2} \text{ d}^{-1}$ is the Delhi reference value, and $SHGC = \text{clip}(1 - 0.05 U_{win}, 0.20, 0.65)$.

At each timestep, heating and cooling demands are calculated as the energy required to maintain the zone within the setpoint band:

$$Q_{heat}(t) = \max(0, C_{eff}(T_{heat,sp} - T_m^*)/\Delta t) \quad (14)$$

$$Q_{cool}(t) = \max(0, C_{eff}(T_m^* - T_{cool,sp})/\Delta t) \quad (15)$$

where $T_{heat,sp}$ and $T_{cool,sp}$ are the per-city heating and cooling setpoints and T_m^* is the unconstrained temperature update from (9). Cooling electrical consumption follows from the building's coefficient of performance:

$$Q_{cool,elec}(t) = Q_{cool}(t) / \text{COP} \quad (16)$$

End-use lighting and appliance loads are generated from 24-hour diversity profiles calibrated separately for residential and commercial archetypes, accounting for weekday-weekend variation. Annual aggregates for all four components are summed to yield Q_{total} (kWh), from which EUI ($\text{kWh m}^{-2} \text{ yr}^{-1}$) is computed by dividing by gross floor area $A_{footprint} \times N_{floors}$.

Envelope U-values and infiltration rates are assigned from a four-band construction vintage table calibrated to Indian building practice, with small multiplicative Gaussian noise ($\sigma = 8\%$ for thermal, $\sigma = 5\%$ for glazing) to prevent degenerate discrete clustering in the training targets. The RC model is intentionally simplified relative to multi-zone EnergyPlus simulations; its role is to produce physically ordered, internally consistent training labels at the batch scale required for surrogate training, not to match individual metered bills.

D. Surrogate Model Training

1) *Train-Test Split*: For each city, the 5,000 RC-simulated buildings are split into 4,000 training and 1,000 test records using a spatial split: records whose centroid latitude falls below the 20th percentile of the city footprint latitude range are assigned to the test set, preventing spatial autocorrelation leakage between training and evaluation [19].

2) *Model Candidates*: Four base estimators are evaluated: Random Forest (RF, 350 trees), XGBoost (350 trees, learning rate 0.05, max depth 6, subsampling 0.8), LightGBM (350 leaves 63, subsampling 0.8), and a Multi-Layer Perceptron (MLP, hidden sizes 128-64-32, ReLU, early stopping). The best single-model predictor is selected on test-set R^2 , then combined with the other tree-based models in a Stacking Ensemble whose meta-learner is a Ridge regressor with 5-fold cross-validation. All feature preprocessing uses median imputation of missing values followed by standard scaling for the MLP. Quantile Gradient Boosting at the P10 and P90 levels provides prediction intervals.

3) *Ablation Study Design*: Three progressive feature subsets are evaluated per city:

- **Tier I** (9 features): Class 1 satellite indices only.
- **Tier II** (18 features): Class 1 + Class 2 morphology.
- **Tier III** (21 features): Class 1 + Class 2 + Class 3 climate.

4) *Leave-One-City-Out Experiment*: For each of the four cities in turn, the remaining three cities' 5,000-building datasets are pooled as training data (15,000 buildings) and the held-out city's 5,000 records serve as the test set. The LOCO metric directly measures cross-zone generalisation without any re-calibration on target-city data.

5) *Pooled Model*: A fifth surrogate is trained on all four cities combined (20,000 buildings), using a stratified city-proportional spatial split. This pooled model is evaluated on the combined 4,000-record held-out set and is subsequently used for 100k inference across all cities.

VI. RESULTS

A. Physics Simulation Output

Table III summarises the annual EUI distributions from the RC simulator for 5,000 buildings per city, disaggregated by building type.

TABLE III: Physics Simulation: Mean EUI by City and Building Type

City	Zone	EUI ($\text{kWh/m}^2/\text{yr}$)		BEE EPI Office (target)
		Residential	Commercial	
Delhi NCR	Composite	207.7	273.8	179
Ahmedabad	Hot & Dry	229.4	313.0	173
Kolkata	Warm & Humid	203.2	271.7	182
Bengaluru	Temperate	172.9	249.2	155

BEE EPI target represents the benchmark for efficient office buildings. Simulated EUI represents the full stock average including a range of construction vintages and occupancy rates; the gap to BEE EPI is consistent with the stock-mean/benchmark distinction documented in [2].

The climate gradient is clearly visible in the end-use breakdown (Table IV). Ahmedabad records the highest cooling demand (mean 44,632 kWh) driven by CDD = 3,200, while Bengaluru produces essentially zero heating (HDD = 100, entirely satisfied by passive means in the model). Kolkata's persistent humidity elevates infiltration-driven cooling loads despite relatively moderate temperatures.

TABLE IV: End-Use Energy Breakdown by City (Mean Annual kWh/building)

City	Heating	Cooling	Lighting	Appliances
Delhi NCR	618	38,579	2,122	5,531
Ahmedabad	32	44,632	1,860	5,399
Kolkata	132	33,300	1,677	4,361
Bengaluru	≈ 0	34,651	2,499	5,995

B. Ablation Study

Table V presents R^2 and MAPE for each feature tier across cities using the 5,000-building-per-city setup (4,000 train, 1,000 test per city).

TABLE V: Ablation Study: R^2 by Feature Tier and City (5k setup)

City	Tier	R^2	MAPE (%)	RMSE (kWh)	n
Delhi NCR	Tier I (C1)	-0.052	75.8	47,722	1,000
	Tier II (C1+C2)	0.962	9.6	9,124	1,000
	Tier III (C1+C2+C3)	0.929	9.7	12,422	1,000
Ahmedabad	Tier I	0.020	68.8	67,582	1,000
	Tier II	0.795	10.0	30,943	1,000
	Tier III	0.787	10.2	31,496	1,000
Kolkata	Tier I	0.223	58.3	41,286	1,000
	Tier II	0.920	9.7	13,248	1,000
	Tier III	0.921	9.8	13,206	1,000
Bengaluru	Tier I	-0.549	66.6	52,383	1,000
	Tier II	0.922	10.2	11,728	1,000
	Tier III	0.915	10.4	12,240	1,000
Pooled (4 cities)	Tier I	0.002	69.1	52,314	4,000
	Tier II	0.858	12.3	19,744	4,000
	Tier III	0.877	9.8	18,341	4,000

Three observations stand out. First, satellite spectral indices alone are insufficient predictors of building EUI at any tier ($R^2 = -0.549$ to 0.223 , MAPE 58–76%); the null hypothesis that spectral imagery directly encodes building energy cannot be sustained. Second, adding building-morphology features (Tier II) produces a step-change improvement of roughly 0.70 to 1.47 in R^2 , driven primarily by footprint area, which exhibits a Pearson correlation of $r = 0.889$ with annual total energy (kWh) because nearly every modelled end-use scales with conditioned floor area. Third, city-level climate features (Tier III) provide a marginal further improvement in the pooled model ($R^2 : 0.858 \rightarrow 0.877$) but are neutral or slightly negative within individual cities where HDD/CDD are constant and carry no within-city discrimination.

This ablation finding has a direct design implication: pipelines that lack building-footprint area information—relying solely on spectral imagery—will not achieve predictive accuracy adequate for planning purposes regardless of model complexity.

C. Per-City Surrogate Performance

Table VI reports the best per-city model selected from the candidate set on the 1,000-building spatial test set.

Kolkata and Bengaluru return the highest within-city R^2 values (0.969, 0.963), reflecting tighter EUI variance driven by their more uniform climate regimes. Ahmedabad,

TABLE VI: Per-City Surrogate Model Performance on Spatial Test Set

City	Zone	Best Model	R^2	RMSE (kWh)	MAE (kWh)	MAPE (%)
Delhi NCR	Composite	Stacking Ensemble	0.941	—	—	10.8
Ahmedabad	Hot & Dry	Random Forest	0.889	—	—	10.9
Kolkata	Warm & Humid	XGBoost	0.969	—	—	10.1
Bengaluru	Temperate	XGBoost	0.963	—	—	10.9
Pooled (4 cities)	—	Stacking Ensemble	0.971	6,149	3,273	11.8

All models trained on 4,000 buildings (spatial split); test set $n = 1,000$ per city. Pooled test $n = 4,000$.

with the widest commercial-to-residential EUI spread (229–313 kWh m⁻²), is the most challenging single-city task ($R^2 = 0.889$). The pooled model ($R^2 = 0.971$) benefits from the larger training corpus and climate-feature encoding of inter-city variation.

D. SHAP Feature Importance

TreeSHAP analysis on the Ahmedabad Random Forest model (Fig. 2) reveals the dominant role of building morphology. The three highest-importance features are footprint_area_m2 (mean |SHAP| = 25,098 kWh), n_floors (5,326 kWh), and height_m (5,325 kWh)—all Class 2 morphological features. building_type is the highest-ranked Class 1 feature (3,145 kWh). All six pure spectral indices (NDVI, NDBI, ENDISI, MNDWI, UI, BSI) individually contribute less than 320 kWh, collectively accounting for roughly 4.6% of the total mean absolute SHAP value. The three Class 3 climate features (CDD, HDD, RH) contribute approximately 330 kWh combined. This hierarchy is consistent across the other cities and with the ablation R^2 pattern.

E. Leave-One-City-Out Generalisation

Table VII presents LOCO results. The in-city test R^2 column provides the reference against which LOCO degradation is measured.

TABLE VII: Leave-One-City-Out (LOCO) Cross-Zone Generalisation

Left-Out City	Zone	In-City R^2	LOCO R^2	RMSE (kWh)	MAPE (%)
Ahmedabad	Hot & Dry	0.889	0.900	19,545	13.1
Delhi NCR	Composite	0.941	0.861	21,471	11.2
Kolkata	Warm & Humid	0.969	0.819	21,863	11.8
Bengaluru	Temperate	0.963	0.628	36,111	21.0

Training set: 15,000 buildings from the three non-held-out cities. Test set: 5,000 buildings from the left-out city.

Ahmedabad generalises remarkably well: its Hot & Dry climate shares building morphology distributions with Delhi and Kolkata, and the LOCO R^2 of 0.900 actually exceeds the in-city estimate (0.889), suggesting the three-city pooled model captures the variance more efficiently than the 4,000-sample single-city model. Delhi and Kolkata exhibit moderate degradation (LOCO $\Delta R^2 \approx -0.08$ to -0.15), likely reflecting the degree-day mismatch between the training set and the target city’s thermal regime. Bengaluru is the clearest outlier: trained on three cities with CDD $\geq 2,900$, the model encounters a city whose CDD is only 700 and whose heating demand is

Figures/shap_bar_ahmedabad.png

Fig. 2: Mean absolute SHAP values for Ahmedabad (Random Forest, 21 features). Features are grouped by class: Class 2 morphology dominates, Class 1 spectral and Class 3 climate features contribute marginally.

effectively zero. The resulting LOCO R^2 of 0.628 and MAPE of 21.0% indicate that the Temperate zone requires dedicated calibration samples for acceptable deployment accuracy.

F. City-Scale Inference at 100k Buildings

Table VIII summarises the 100k deployment results. Prediction intervals are the P10–P90 quantile regressor band.

The EUI gradient across cities is physically consistent with the climate configuration: Ahmedabad (Hot & Dry, highest CDD) records the highest mean EUI (241.3 kWh m^{-2}), while Bengaluru (Temperate, $\text{CDD} = 700$) records the lowest (194.6 kWh m^{-2}). Delhi and Kolkata occupy intermediate positions (227.6 and 221.9 kWh m^{-2} respectively). The wider P10–P90 interval for Ahmedabad (96.5 kWh m^{-2}) compared to Bengaluru (77.9 kWh m^{-2}) reflects greater EUI variance in the Hot & Dry zone, where commercial-to-residential EUI contrast is most pronounced.

Computational speedups range from $2,942\times$ to $18,193\times$ relative to the extrapolated RC simulation time. The lowest speedup for Ahmedabad arises because its surrogate (Random Forest) has a larger fitted model size than the XGBoost models used for Kolkata and Bengaluru, reflecting that a larger model is necessary to capture Ahmedabad’s wider variance structure.

G. Carbon Intensity Estimation

Using city-specific grid emission factors (Table I), the predicted building electricity consumption is converted to annual CO_2 -equivalent emissions:

$$E_{\text{CO}_2} = Q_{\text{total}} \cdot \text{EF}_{\text{grid}} \quad (17)$$

At 100k-building scale, Kolkata produces the highest carbon intensity ($\text{EF} = 0.94 \text{ kgCO}_2/\text{kWh}$) despite its lower mean EUI than Ahmedabad, underscoring that grid decarbonisation has a stronger near-term effect on building-sector emissions in the Eastern region than envelope efficiency improvements alone.

VII. DISCUSSION

A. Physics-Informed Ground Truth: Scope and Limitations

The surrogate models in this paper are trained against RC-simulator-generated labels rather than measured utility bills. This is an acknowledged limitation that is common to the broader class of physics-informed surrogate studies. The practical consequence is that absolute EUI predictions reflect the assumptions embedded in the RC model—notably, that mechanical cooling is active for all buildings across the full CDD season and that occupancy follows the specified stochastic prior. For Indian residential buildings, where natural ventilation and evaporative cooling are widespread, this assumption inflates the simulated cooling demand relative to what metered data would show. The gap between simulated residential EUI ($173\text{--}229 \text{ kWh m}^{-2}$) and the BEE EPI reference for efficient residential buildings (45 kWh m^{-2}) should therefore be understood as the difference between a mechanical-cooling scenario EUI and a best-practice benchmark, not as model error.

For commercial buildings the simulated EUI ($249\text{--}313 \text{ kWh m}^{-2}$) exceeds the BEE office EPI by a factor of 1.5–1.8, a range that is consistent with published observations of typical Indian commercial stock performance relative to code targets [2]. This suggests that the commercial segment of the model is reasonably well calibrated in relative terms.

B. The Role of Building Morphology

The ablation and SHAP analyses converge on a clear message: footprint area is the single strongest predictor of annual building energy because all modelled end-uses scale with conditioned floor area. This is not a failure of the spectral feature set but a fundamental physical relationship. Consequently, the GeoAI pipeline’s primary value lies not in the spectral indices per se but in its ability to locate buildings from satellite data and associate them with GOB footprint areas at scale. The spectral indices contribute three secondary functions: classifying buildings as residential or commercial (*building_type*), modulating stochastic priors for age and WWR through NDBI-correlated boosts, and enabling LOCO transfer through climate-zone-discriminating patterns in urban density indices. Future work should examine whether very-high-resolution imagery ($\leq 0.5 \text{ m}$) can provide independent height estimates via roof-shadow length analysis [15], which would reduce the uncertainty associated with the probabilistic floor-count prior.

C. Implications of the LOCO Experiment

The LOCO result for Bengaluru ($R^2 = 0.628$) is the most operationally significant finding of the study. It indicates that deploying a surrogate trained on high-CDD cities to

TABLE VIII: City-Scale Surrogate Inference: 100,000 Real GOB Buildings per City

City	Zone	Buildings	Mean EUI (kWh/m ²)	Median EUI (kWh/m ²)	Total GWh	P10 EUI (kWh/m ²)	P90 EUI (kWh/m ²)	Speedup (vs RC)
Delhi NCR	Composite	99,964	227.6	208.3	4,739 GWh	182.1	273.1	3,836×
Ahmedabad	Hot & Dry	99,991	241.3	223.7	5,300 GWh	193.0	289.5	2,942×
Kolkata	Warm & Humid	99,997	221.9	206.3	3,969 GWh	177.5	266.3	18,193×
Bengaluru	Temperate	99,994	194.6	176.7	4,290 GWh	155.7	233.6	17,021×

Speedup computed as extrapolated RC simulation time (5k rate $\times$ 20) divided by surrogate inference wall-clock time. Mean EUI and total GWh reflect predictions on real GOB footprints, not the 5k training set.

a temperate city without any local calibration data would produce predictions with MAPE $\approx$ 21%, roughly twice the within-city error. Practitioners considering deployment in data-scarce temperate cities should treat the surrogate as providing order-of-magnitude estimates only, and should budget for a minimum calibration sample of several hundred locally simulated buildings to re-anchor the model. The strong LOCO performance for Ahmedabad ($R^2 = 0.900$) has a positive implication: Hot & Dry zone cities that lack local GOB density for training (e.g., Jodhpur, Bikaner) could leverage a surrogate trained on Delhi and Kolkata with acceptable accuracy.

D. Tier I Satellite-Only Prediction

The finding that Tier I models explain only 11–15% of variance per city is worth interpreting carefully. It does not imply that Sentinel-2 spectral data are uninformative for building energy applications; rather, it reflects that the specific energy target used here (total annual kWh) is dominated by floor area, and spectral indices contain negligible information about floor area. Tasks where spectral data are more directly informative—such as roof type classification, surface albedo estimation, or vegetation fraction—remain viable Tier I applications. The Tier I architecture is further constrained by the point-sampling methodology, which samples a single pixel at the building centroid; polygon-level zonal aggregation over the full footprint would likely recover more spectral variance and should be evaluated in future iterations.

VIII. CONCLUSION

This paper described a fully reproducible GeoAI-surrogate pipeline for urban building energy assessment, applied to four Indian cities spanning four ECBC climate zones. The principal findings can be stated as follows.

Building-morphology features—footprint area in particular—dominate EUI prediction; satellite spectral indices alone are insufficient, explaining fewer than 15% of the variance in annual energy consumption per building. This finding provides an empirical basis for pipeline design: surrogate models that lack building-footprint information will not reach planning-grade accuracy regardless of spectral complexity.

Cross-city generalisation is strong for Hot & Dry, Composite, and Warm & Humid zones (LOCO $R^2 = 0.819$ – 0.900) and degrades significantly for the Temperate zone ($R^2 = 0.628$), which requires dedicated local calibration because its

mechanical-conditioning demand structure is qualitatively different from the other three zones.

City-scale inference at 100,000 buildings per city is completed in under one second per city (2,942–18,193 $\times$ faster than the underlying RC simulator), making the surrogate framework well suited to iterative planning scenarios.

The methodology operates entirely on open data—Google Open Buildings, Sentinel-2 via GEE, and EPW files—without requiring measured utility data, making it transferable to any Indian city for which an ECBC climate zone designation and EPW file are available.

A. Future Directions

Several extensions would substantially strengthen the framework. Acquiring even a small metered calibration dataset (50–200 buildings per city) from DISCOM feeder data or building audit records would enable absolute EUI calibration and shift the validation from physics-to-surrogate to surrogate-to-measured. SHAP analysis should be extended across all four cities to assess whether feature importance ranks vary with climate zone. The floor-count estimation could be improved by incorporating satellite-derived building height from stereo imagery or OpenStreetMap height tags. Finally, extending the target from annual total energy to monthly profiles would enable peak-demand and grid-planning applications.

APPENDIX

The pipeline is implemented as five Python scripts (Scripts/01–05) using a shared `city_config.py` configuration module. All scripts accept a `-city` argument; passing `all` executes the full multi-city run. Random seed 42 is set globally. Key output artefacts include per-city feature CSVs (`01_features_{city}.csv`), physics ground truth CSVs (`02_physics_ground_truth_{city}.csv`), ablation results (`04_ablation_{city}.csv`), LOCO results (`Results/multi_city/04_loco_results.csv`), trained model files (`Models/multi_city/{city}/best_model.j`) and 100k inference predictions (`05_inference_{city}_100k.csv`).

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